Portfolio Deliverable 1: Multi-Omics Target Discovery Package

Disease: Ovarian Cancer

1. Introduction: Disease and Question

Ovarian cancer remains a major clinical challenge because it is often diagnosed at an advanced stage. High recurrence rates and limited long-term treatment responses further complicate disease management. Identifying promising therapeutic targets requires more than high gene expression or mutation frequency. A strong candidate should have consistent multi-omics evidence, biological relevance, and therapeutic potential. This study integrates transcriptomic, proteomic, and genomic data to prioritize ovarian cancer targets. Protein language models are then used to predict druggability. AlphaFold structural analysis further evaluates the potential for small-molecule targeting in selected proteins. Together, these approaches help identify candidates that are both biologically relevant and potentially actionable.

2. Multi-Omics Evidence Integration

Module 2 loaded 20,032 RNA rows, 6,475 protein rows and 2,205 mutation rows. Gene-symbol harmonization yielded 751 genes present in all three layers. For each shared gene, the transcriptomic and proteomic summaries captured abundance, while the genomic feature represented mutation frequency. Each layer was rank-percentile normalized, then combined with equal one-third weights. This balances numerical scales but does not imply that the layers are equally informative biologically.

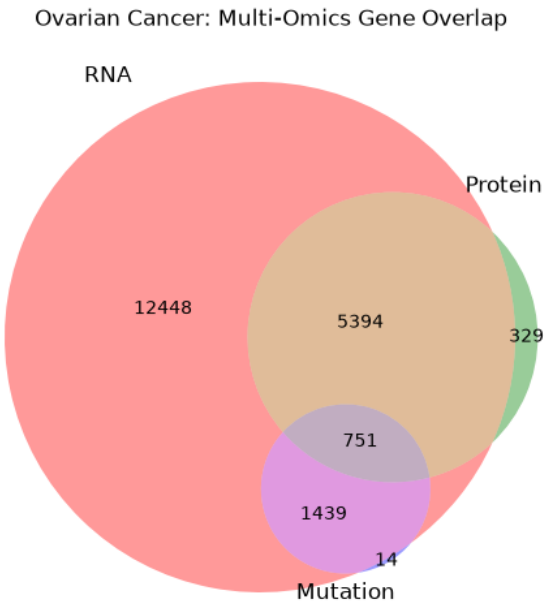


Figure 1. Overlap of unique gene identifiers across ovarian cancer transcriptomic, proteomic, and genomic datasets. The central intersection of 751 genes defines the set used for multi-omics integration. Dataset row counts may differ

slightly from unique gene counts.

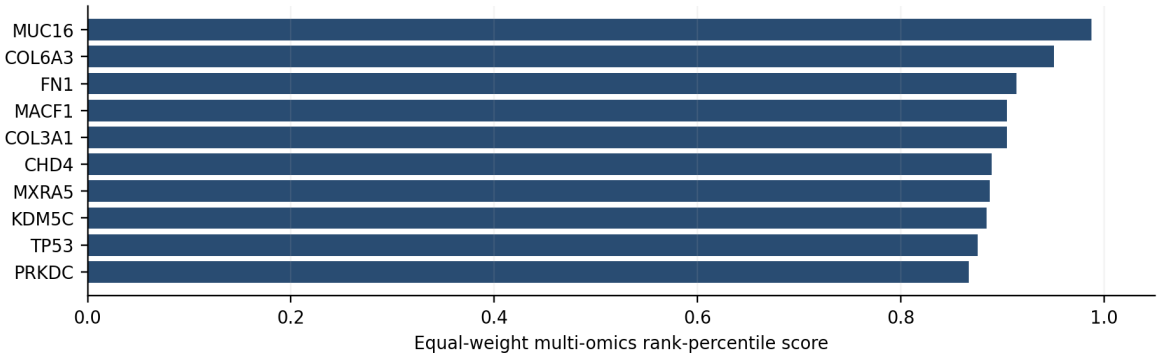


Figure 2. Top ten Module 2 targets by equal-weight integrated score. The bars show relative ranking within the 751 shared genes, not differential expression or causal importance

Among 54 samples matched between RNA and protein matrices, median gene-wise Spearman correlation was 0.45 (range −0.48 to 0.88); 725/751 genes had positive correlations. CDH6 was most coupled (ρ = 0.88), whereas ATP1A2 was most discordant (ρ = −0.48). These correlations reflect cross-sample RNA–protein co-variation, not agreement in tumor-versus-normal fold changes.

3. Sequence-based tractability and re-ranking

Module 3 mapped the 15 highest Module 2 genes to reviewed proteins and used ESM-C 300M embeddings to estimate two distinct clinical-precedence priors: small-molecule tractability (p_sm) and antibody tractability (p_ab). Four target proteins overlapping the training panel were excluded to prevent direct label leakage. Family-aware five-fold validation yielded AUROC 0.741 and average precision 0.548, versus 0.840 and 0.717 for a random split (chance AP 0.332). The gap suggests related-protein leakage makes random-split performance optimistic.

Re-ranking used 0.60 × percentile(Module 2 score) + 0.40 × percentile(max[p_sm, p_ab]) within the 15 candidates. The 60:40 weighting gives more importance to the multi-omics evidence while still accounting for predicted druggability. The resulting score is used to prioritize candidates rather than predict clinical success.

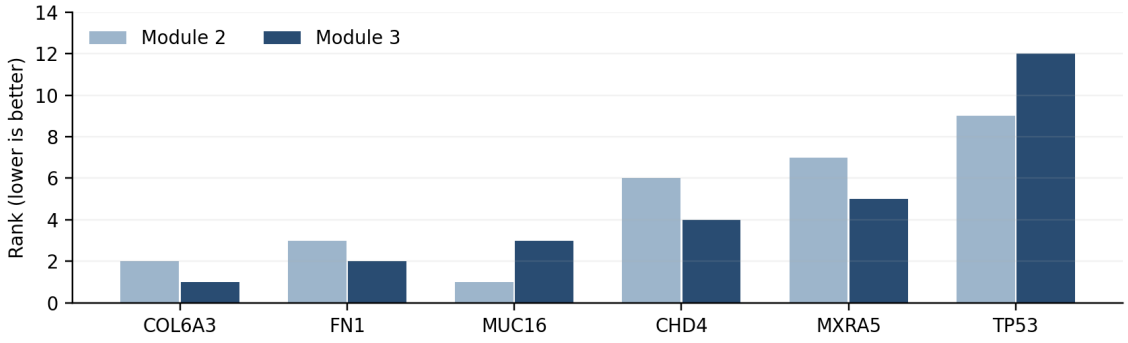


Figure 3. Changes in shortlist position after adding sequence-based tractability. Lower numerical rank indicates higher priority; the Module 3 ordering is restricted to the 15 screened genes

Table 1. Re-ranking of prioritized ovarian cancer targets from Module 2 (multi-omics evidence) to Module 3 (protein language model-based druggability assessment)

Gene	M2 → M3	p_sm	p_ab	Interpretation
COL6A3	2 → 1	0.835	0.212	High evidence and model prior; long-sequence caveat
FN1	3 → 2	0.497	0.291	Consistent RNA/protein evidence; extracellular target
MUC16	1 → 3	0.000	0.149	Highest multi-omics evidence; lower modeled tractability
CHD4	6 → 4	0.684	0.000	Small-molecule-oriented follow-up
MXRA5	7 → 5	0.005	0.281	Relatively stronger antibody-oriented prior
TP53	9 → 12	0.033	0.011	Biological anchor; low predicted direct tractability

COL6A3 became the leading sequence-triaged candidate, while MUC16 fell despite strong disease-layer evidence. The priors encode known development patterns rather than experimentally demonstrated binding. Nine of 15 proteins exceed the 2,046-residue embedding input limit; uncertain sequence coverage particularly tempers confidence in long proteins such as COL6A3 and FN1.

4. Structural confidence and binding-pocket assessment

Module 4 selected KRAS (Module 2 rank 41) and PIK3CA (rank 514) for a structure-focused comparison. They were not Module 3 top-ranked discovery leads, and the structural assessment should not be represented as validation of COL6A3 or FN1. AlphaFold structures were compared with experimental structures 6OIM (KRAS) and 4JPS (PIK3CA), and fpocket candidates were checked against model confidence and ligand-bound sites.

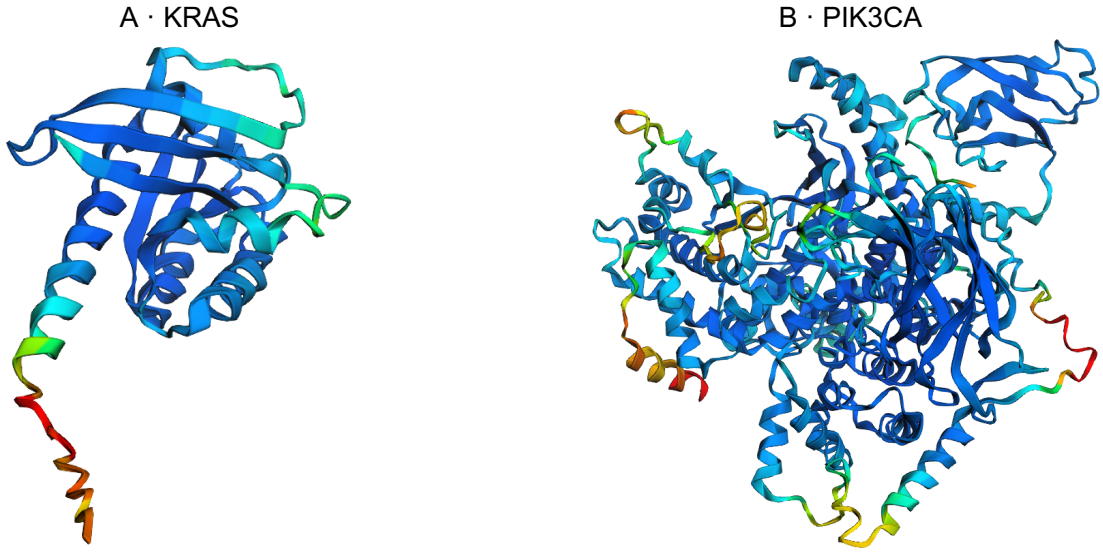


Figure 4. AlphaFold-predicted KRAS (A) and PIK3CA (B), colored by residue-level confidence (pLDDT). Blue/cyan regions are more confident than yellow/orange/red segments. These images show model confidence, not pocket locations

Table 2. Comparative structural assessment of KRAS and PIK3CA using AlphaFold predictions, experimental PDB structures, and fpocket analysis

Metric	KRAS	PIK3CA
Mean pLDDT	91.5	92.4
% residues pLDDT < 50	2.1%	1.1%
Confident-residue RMSD to PDB	2.19 Å (6OIM)	0.90 Å (4JPS)
Top kept fpocket score	0.612 (pocket 1)	0.261 (pocket 85)
Top pocket vs bound ligand	78% of GDP site; 29% of MOV site	100% of 1LT site
Structural conclusion	Possible but uncertain	Small-molecule pocket supported

KRAS illustrates a critical failure mode. Its highest-ranked pocket primarily matches the GDP nucleotide site, not the sotorasib-bound switch-II pocket (MOV). The latter is best covered by a much lower-ranked pocket (rank 10; 38% coverage), consistent with cryptic, ligand-dependent pocket formation. In contrast, PIK3CA pocket 85 fully overlaps the experimentally observed 1LT ligand-binding site in PDB 4JPS. High global pLDDT alone does not establish pocket druggability.

5. Prioritized shortlist and decision rationale

Table 3. Final prioritized ovarian cancer target shortlist integrating multi-omics evidence, protein language model-based druggability predictions and structural assessment

Priority	Target	Evidence / tractability	Next decision
1 Discovery	COL6A3	M2 #2 → M3 #1; p_sm 0.835; final 0.960	Confirm full-sequence embedding; tumor-specific dependency and accessibility
2 Discovery	FN1	M2 #3 → M3 #2; ρ 0.714; final 0.867	Test tumor/stromal source, selectivity and accessible epitope
3 Discovery	MUC16	M2 #1 → M3 #3; final 0.840	Assess surface localization, antigen heterogeneity and antibody strategy
4 Discovery	CHD4	M2 #6 → M3 #4; p_sm 0.684	Examine ATPase-site selectivity and functional dependency
Structural comparator	PIK3CA	M2 #514; pLDDT 92.4; 100% ligand-site coverage	Assess mutation/context-specific ovarian-cancer relevance
Structural caution	KRAS	M2 #41; high-confidence fold but cryptic switch-II site	Use mutant/ligand-specific conformations; do not trust top pocket alone

The final shortlist (Table 3) prioritizes candidates based on multi-omics evidence and sequence-derived tractability while retaining structural findings as a separate line of evidence. COL6A3 emerged as the leading discovery candidate, advancing from Module 2 rank 2 to Module 3 rank 1 with a final score of 0.960 and a small-molecule prior of 0.835. However, its large protein size

and uncertain embedding coverage warrant further investigation before interpreting this score as evidence of direct targetability. FN1 ranked second, supported by strong RNA–protein concordance (ρ = 0.714), although its extracellular localization raises questions about tumor specificity and accessibility. MUC16 retained the strongest initial multi-omics evidence but fell to third after incorporating tractability predictions, suggesting that an antibody-oriented strategy may be more appropriate to investigate than conventional small-molecule targeting. CHD4 ranked fourth, with a relatively high small-molecule prior (0.684), making it a candidate for functional dependency and selectivity studies.

KRAS and PIK3CA were evaluated separately as structural case studies rather than ranked alongside these discovery candidates. PIK3CA demonstrated strong ligand-site coverage, whereas the KRAS analysis highlighted the limitations of relying on the highest-scoring predicted pocket without considering ligand-specific conformations. Together, these findings support a prioritized discovery shortlist while identifying the additional evidence needed before target advancement.

6. Limitations and Validation

Several limitations prevent the current findings from establishing therapeutic efficacy or clinical relevance. The multi-omics analysis was limited to 751 genes shared across three datasets, and RNA-protein concordance reflects correlation rather than differential expression or causal disease involvement. Protein language model-based druggability scores are predictive rather than experimentally validated, with uncertainty for proteins exceeding the model's sequence-length limit. Furthermore, structural analysis was performed only for KRAS and PIK3CA, so structural tractability cannot be inferred for the highest-ranked discovery candidates. TP53 illustrates the distinction between disease relevance and direct druggability. Despite its high mutation frequency (0.828), it fell from Module 2 rank 9 to Module 3 rank 12 because of low predicted therapeutic priors. ATP1A2 also exhibited substantial RNA–protein discordance (ρ = −0.48), demonstrating the limitations of relying on transcript abundance alone.

Experimental validation should prioritize COL6A3, the highest-ranked discovery candidate. First, its expression should be confirmed in independent ovarian tumor cohorts and compared with matched normal tissues using RNA-seq and immunohistochemistry. Spatial localization should distinguish expression by malignant cells from contributions by cancer-associated fibroblasts and other stromal populations. Second, CRISPR-mediated perturbation or RNA interference in ovarian cancer cell lines and patient-derived organoids should assess effects on proliferation,

viability, and invasion, with appropriate non-targeting controls and rescue experiments to establish specificity. Because COL6A3 is an extracellular matrix protein, functional studies should also consider stromal co-culture models rather than relying exclusively on tumor-cell monocultures. These experiments would help determine whether COL6A3 contributes to ovarian cancer progression and whether targeting it could have therapeutic value. Further studies would then be needed to assess its safety and suitability for drug development.

7. References

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